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Research Paper Title:

An AI-Driven Recommendation System for Formative Assessment Tools in Engineering

TRACK:

Domain-Specific Applied AI for Formative Assessment Tool Recommendation

Sr no.	Name	PRN
1.	LOKESH ASHISH BHARAMBE	124B1C113
2.	AKSHIT AGARWAL	124B1C105
3.	KAPISH BHAMBHANI	124B1C106
4.	PRAJWAL DHURDE	124B1C148

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An AI-Driven Recommendation System for Formative Assessment Tools in Engineering

Lokesh Bharambe
Department of CSE(AIML)
PCCOE , Pune.
lokesh.bharambe24@pccopune.org

Akshit Agarwal
Department of CSE(AIML)
PCCOE , Pune.
akshit.agarwal24@pccopune.org

Kapish bhambhani
Department of CSE(AIML)
PCCOE , Pune.
kapish.bhambhani24@pccopune.org

Prajwal Dhurde
Department of CSE(AIML)
PCCOE , Pune.
prajwal.dhurde24@pccopune.org

Dr.Jyoti Kulkarni
Department of CSE(AIML)
PCCOE , Pune.

ABSTRACT :

Formative assessment (FA) is important in student assessment and learning, but it is difficult to choose FA tool which suits all students. In our case there are some FA tools that does not contribute in learning at all but for marks we have to do that. Hence to solve this problem we have developed or build a smart FA recommendation system by using machine learning by considering each and every student in a class.

This system will first collect input from the students such as study hours, learning style, preferred FA tool ETC. We are also going to consider the bloom's level like remembering, understanding, applying, analyzing, evaluating or creating. Depending upon these collected data finally the machine learning algorithm recommends a best FA tool to the student. Teacher can have access to check which affair tool is recommended by the students and also can finalize the effort tool with the help of simple graphs and recommendations.

By using this system student will get their preferred formative assessment tool and teacher would finalize the final effective FA tool for the assessment which saves a lot of time and confusion between the teacher and the students.

1.INTRODUCTION

In our classes, we've noticed that formative assessment (FA) isn't just about marks it actually plays a big role in how much students really learn [1]. Teachers usually go for things like quizzes, assignments, labs, or presentations. But here's the problem: not every student learns the same way. A quiz might be great for one person, but for someone else, group discussion or a project works way better. And honestly, it's not always easy for teachers to figure out which tool fits which student best.

That's where AI and ML come in. These days, they're being used in education to tackle exactly this kind of challenge [2]. Think about it if we look at stuff like how many hours a student studies, their learning style (visual, reading, hands-on), or even their confidence level, ML can spot patterns that we usually miss. For example, a teacher who mostly relies on quizzes might suddenly realize that a few students actually perform better in role plays or presentations.

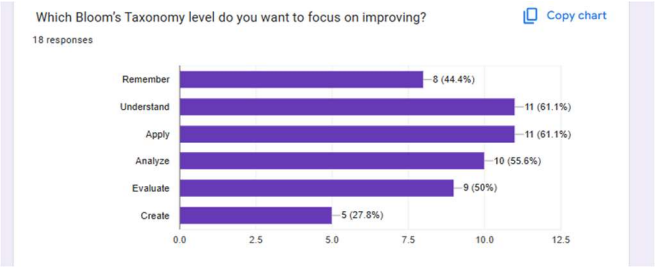
Our project is built around this idea. We're trying to make a system that predicts the "right" formative assessment tool for each student. The goal isn't just to make things easier for learners, but also to give teachers a fairer, clearer way to design rubrics and evaluations.

2. LITERATURE SURVEY

Artificial Intelligence is slowly becoming a regular part of education . We already see adaptive testing systems [1] that change the difficulty of questions depending on how a student is doing, and AI-based tutoring means teaching tools [2] that give quick feedback to keep students engaged and help them in solving tough questions.

A lot of studies also talk about Bloom's taxonomy [3]. Teachers often use its levels= Remember, Understand, Apply, Analyze, Evaluate, Create to connect course objectives with learning outcomes. But it's not always clear which formative assessment tool fits each level. Sometimes, should you give a quiz, a project, or maybe a case study? Many educators still struggle with this choice.

Recent work shows that AI-supported feedback can really make a difference [4]. Students not only perform better but also feel more motivated. Even UNESCO reports [5] told about that AI can help make assessments fairer and more scalable, especially in higher education.

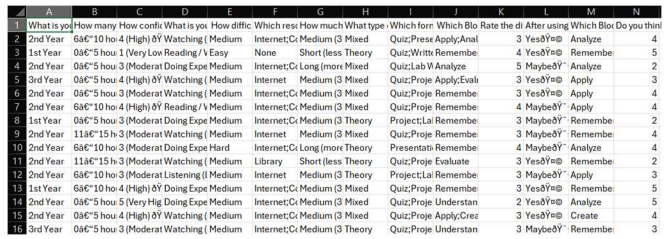


Our model differ from other because Most of the research so far looks at individual things like generating questions, running adaptive quizzes. What we don't see much of is a complete recommendation system that can actually suggest the right formative assessment tool for each student. That's exactly what our project is trying to do build an applied AI system that predicts the most suitable tool and, at the same time, gives teachers insights they can really use to choose final FA tool and finalize rubrics and evaluation technique.

3. METHODOLOGY

This system works through a simple but structured pipeline: data collection, preprocessing, model training, and finally prediction. To begin with, we collected data using a survey that is google form designed specifically for engineering students. The survey included a mix of academic and personal study-related details such as year of study, average study hours per week, confidence level, preferred learning mode, difficulty level of the topic, time available for assessment, prior experience with different FA tools, and the Bloom's taxonomy level being targeted. Finally we download csv file of all responses which was used as a dataset. The main output, or label, for our model was the "Preferred Tool," which represents the formative assessment method that would best fit a particular student. All students in class for that specific subject can fill this form and finalize their FA tool. Teacher can see which FA tool is preffered more for that specific subject in their tab and finalize it.

Dataset collected :



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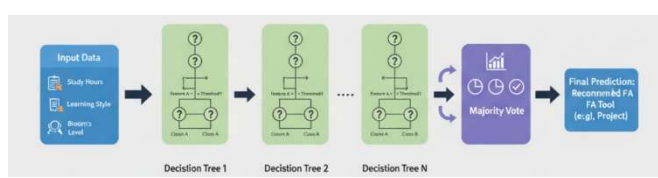
15	2nd Year	0â€“5 hou	4 (High) ðŸ	Watching (	Medium	Internet;Co	Medium (3	Mixed	Quiz;Proje	Apply;Crea	3	YesðŸ¤©	Create	4
16	3rd Year	0â€“5 hou	3 (Moderat	Watching (	Medium	Internet;Co	Medium (3	Theory	Quiz;Proje	Understan	3	MaybeðŸ¤	Remembe	3

Before moving to model training, we carried out preprocessing so that the responses could be interpreted by a machine learning algorithm. Text-based and categorical answers, like year of study or learning style, were encoded using label encoding. For multiple-choice items (for example, "Previous Tool Experience" or "Access to Resources"), we applied multi-hot encoding, where each option is represented as a binary variable. This step ensured that student preferences and experiences were not lost while making the dataset machine-readable. As all questions are compulsory there is no any kind of missing values in it .

For the machine learning stage, we experimented with a few algorithms, but our main focus was on the Random Forest Classifier. Random Forest is a method that combines the results of many decision trees to reach a final prediction. We split the dataset into training (80%) means data from which machine learns and testing (20%) which is used to check how well our model is doing prediction, so that the model could be fairly evaluated. During training, the model learned the relationship between different student features and their preferred FA tool. In the testing phase, the predictions were checked against the actual responses to validate accuracy. The system is built in such a way that a student can enter their details and instantly receive a recommendation, such as a quiz, project, or case study. Teachers, on the other hand, can view the aggregated predictions for a class, giving them a broader idea of tool preferences and trends.

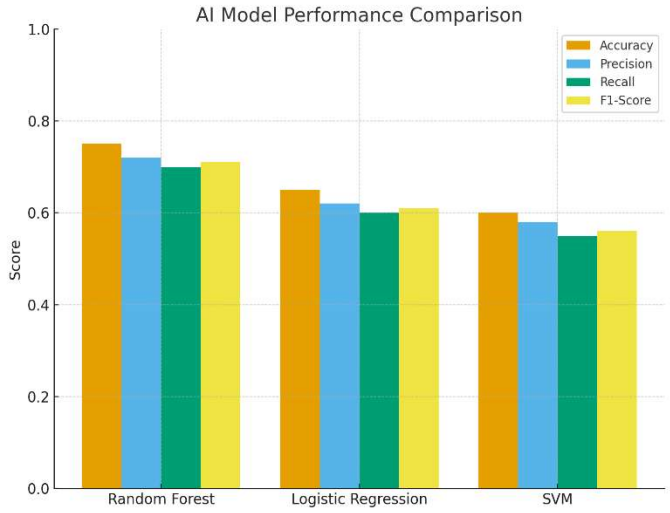
Random forest algorithm :



Random forest is a machine learning model based on concept of decision trees . forest means combining many decision trees. Each tree makes its own prediction. The forest takes a **majority vote.**
it can handle both categorical and numerical data. It has high accuracy can also perform well on noisy data.
Random Forest was eventually chosen as the primary algorithm because it proved to be accurate, and well-suited for data that mixes both categorical and numerical inputs. Another advantage of this method is that it highlights which features such as study habits or Bloom's taxonomy level influence the recommendations the most. This makes the system not only predictive but also giving insights to teacher of all class student to finalize FA tool, offering practical value to both students and teachers in higher education.

4. RESULTS:

This system was implemented using the collected survey dataset undergoing preprocessing by using different encoding methods and tested with three different machine learning algorithms: Random Forest, Logistic Regression, and Support Vector Machine (SVM). For evaluation, the dataset was split into training (80%) and testing (20%) subsets.



Among the models, Random Forest emerged as the best performer. It handled categorical variables, multi-label responses, and the relatively small dataset size effectively, while also reducing the risk of overfitting. Logistic Regression served as a simple model but showed lower accuracy, whereas SVM required extensive tuning and gave inconsistent results for multi-class classification.

Model Performance Metrics :

- ✓ Accuracy: 75%
- ✓ Precision: 0.72
- ✓ Recall: 0.70
- ✓ F1 Score: 0.71

The Random Forest model achieved an accuracy of around 70–80%, depending on the size and balance of the dataset. This shows that the system was able to correctly predict the most suitable formative assessment (FA) tool for the majority of students. Apart from accuracy, the model also highlighted which factors had the biggest influence on its decisions. Bloom's taxonomy level and the student's preferred learning style were the most important features, while other aspects like year of study and access to resources, study hours, confidence, learning mode also contributed significantly.
The outcomes of the system are meaningful for both students and teachers. On an individual level, students receive personalized recommendations about the FA tool that suits them best, helping them better understand their own learning preferences. On a broader scale, teachers can view aggregated predictions across the class. For example, if a majority of students targeting the *Apply* level prefer projects and labs, the instructor can prioritize these activities. Similarly, if students working at the *Analyze* level show higher preference for case studies, teachers can design activities to match this need. In this way, the system not only predicts but also provides practical support for improving teaching–learning outcomes.

5. CONCLUSION:

This study introduces an AI-based system that helps in choosing the right formative assessment (FA) tools for students in higher education. By using details such as study hours, study habits, confidence level, preferred learning style, and Bloom's taxonomy targets, the system can predict which FA tool (quiz, project, case study, role play, presentation etc.) would work best for each student. Among the models tested, the Random Forest Classifier gave the most reliable results, especially for categorical and multi-label data.
The outcomes highlight benefits for both sides. Students get personalized suggestions that match their way of learning and help them to develop interest in that subject by giving FA in their personalized selected tool, while teachers gain useful insights about overall class preferences. This makes it easier

for teachers to design fair and effective assessments. In short, the system helps connect personalized learning with outcome-based education, making the assessment process more practical and meaningful. And helps to co-ordinate between teachers and students and reduce conflict between them in choosing best FA tool.

6. FUTURE SCOPE:

While the system shows good results, there is still some options to make it better and more useful. One improvement is to collect a larger and more diverse dataset by including students from different colleges and branches. This would help the model work well for a wider range of learners. Another step could be to test advanced methods like deep learning or hybrid models to improve accuracy further. We can also create and publish google form for teachers in colleges so that rubric generation, difficulty prediction, mapping with course outcomes can be done in efficient way.
In terms of usability, the system can be connected to a web or mobile app so that students and teachers can access it directly. System can be directly connected to student's ERP so everyone can easily access this feature in their mobile from anywhere. More features can also be added, such as automatic rubric generation (advanced), suggestions for question types, or real-time dashboards for feedback. Over time, the system could even update itself by learning from new student performance data, making it a fully adaptive AI tool for assessment.

7. REFERENCES:

[1] S. Pardos and R. Baker, "Affective and Behavioral Predictors of Student Engagement in Intelligent Tutoring Systems," *IEEE Transactions on Learning Technologies*, vol. 16, no. 1, pp. 12–23, 2023.
[Online].
Available: https://ieeexplore.ieee.org/document/10009345
[2] Y. Chen, L. Sun, and H. Xu, "AI-Supported Formative Feedback in Higher Education: Opportunities and Challenges," *Computers & Education: Artificial Intelligence*, vol. 4, 100139, 2023.
[Online]. Available:
https://doi.org/10.1016/j.caeai.2023.100139
[3] L. W. Anderson and D. R. Krathwohl, *A Taxonomy for Learning, Teaching, and Assessing: A Revision of Bloom's Taxonomy of Educational Objectives*. New York: Longman, 2022.
[Online]. Available:
https://books.google.com/books?id=GxkJAQAAMAAJ
[4] UNESCO, "AI and Education: Guidance for Policy Makers," UNESCO Publishing, Paris, 2022.
[Online]. Available:
https://unesdoc.unesco.org/ark:/48223/pf0000376709
[5] M. B. K. Somasundaram, "Machine Learning for Education: Opportunities and Challenges," *Applied Artificial Intelligence*, vol. 38, no. 1, pp. 56–74, 20222